

EN2091 Laboratory Practice and Projects

Fundamentals of Computer Organization and Design -

Workshop 3

Task Sheet

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Task 1.

*digital multimeter readings are rounded to 0V and 5V. (if >2.5V → 5V and if <2.5V → 0V)

A	B	Q Before (V)	$\bar{Q}$ Before (V)	Q After (V)	$\bar{Q}$ After (V)
0V	5V	0V	5V	5V	0V
0V	0V	5V	0V	5V	0V
5V	0V	5V	0V	0V	5V
0V	0V	0V	5V	0V	5V
5V	5V	0V	5V	0V	0V

Task 2.

When both inputs are 0, Q doesn't change.
i.e. Q remains in the same value
Q Before. Same happens for $\bar{Q}$.

Task 4.

Select	R/ $\overline{W}$	Data in	Q(V)	Data Out (V)
0V	0V	0V	0V	0V
0V	0V	5V	0V	0V
0V	5V	0V	0V	0V
0V	5V	5V	0V	0V
5V	0V	0V	0V	0V
5V	0V	5V	5V	0V
5V	5V	0V	5V	5V
5V	5V	5V	5V	5V

Task 6.

Word Line	Before Applying Input			After Applying Input		
	B	B'	Data	B	B'	Data
	(V)	(V)	(V)	(V)	(V)	(V)
Write Operation						
0V	5V	0V	0V	5V	0V	0V
5V	5V	0V	0V	5V	0V	5V
0V	0V	5V	5V	0V	5V	5V
5V	0V	5V	5V	0V	5V	0V
Read Operation						
0V	0V	0V	0V	0V	0V	0V
5V	0V	0V	0V	0V	5V	0V
5V	5V	0V	0V	5V	0V	5V
0V	0V	0V	5V	0V	0V	5V
5V	0V	0V	5V	5V	0V	5V

Task 8.

Word Line	Before Applying Input		After Applying Input	
	B	Data	B	Data
	(V)	(V)	(V)	(V)
Write Operation				
0V	5V	0V	5V	0V
5V	5V	0V	5V	3.7V
0V	0V	3.7V	0V	3.7V
5V	0V	3.7V	0V	0V
Read Operation				
0V	0V	0V	0V	0V
5V	0V	0V	0V	0V
5V	5V	0V	5V	3.7V
0V	0V	3.7V	0V	3.7V
5V	0V	3.7V	3.7V	3.7V

Task 9.

The voltage at the 'Data' node gradually decreases over the observed time period.

Task 10.

DRAM stores data (voltage) across a capacitor. Charges stored in capacitors discharge over time. So we have to refresh DRAM periodically to keep the stored data in it for a longer time.

Task 11.

Advantage:- Larger capacitor holds charge for a long time. So the time interval to refresh capacitor is long.

Disadvantage:- Larger capacitor takes long time to charge too. So the speed of the capacitor reduces then.

Task 12.

	SRAM	DRAM
speed	fast	slow
hardware complexity	high	low
memory cell density	low	high
cost per byte	high	low